

File permissions in Linux

Project description

I have accomplished adding and removing permissions for files and directories in the lab activity, Including hidden files for authorization and ownership

Check file and directory details

```
researcher2@06ff4b1cc68d:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 May 20 15:33 .
drwxr-xr-x 3 researcher2 research_team 4096 May 20 15:52 ..
-rw--w---- 1 researcher2 research_team  46 May 20 15:33 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 May 20 15:33 drafts
-rw-rw-r-- 1 researcher2 research_team  46 May 20 15:33 project_k.txt
-rw----- 1 researcher2 research_team  46 May 20 15:33 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 May 20 15:33 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 May 20 15:33 project_t.txt
```

Describe the permissions string

r means read.

w means write.

x means execute.

The first character describes whether it is a directory or not. If d, it is a directory, if hyphen, then not a directory.

From 2nd to 4th character, it describes user permissions.

From 5th to 7th character, it describes group permissions.

From 8th to 10th character, it describes other permissions.

If there is a hyphen, then no permission.

Chmod is used to update these permissions in Linux, like:

u+rw - gives permissions to users for read, write and execute.

g-rwx removes permissions to group for read, write and execute.

Chmod o+w project_k.txt adds permission to others for writing the file.

Chmod g-r project_m.txt removes permission to group for reading the file.

Chmod u-w,g-w .project_x.txt removes writing permissions to groups and users.

Chmod g+r .project_x.txt adds reading permission to group.

Ls -la => lists all the contents and permissions of the files in a directory, including hidden files and directories.

Hidden files have “.” before the file’s name.

Change file permissions

```
researcher2@06ff4b1cc68d:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 May 20 15:33 drafts
-rw-rw-rw- 1 researcher2 research_team  46 May 20 15:33 project_k.txt
-rw-r----- 1 researcher2 research_team  46 May 20 15:33 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 May 20 15:33 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 May 20 15:33 project_t.txt
researcher2@06ff4b1cc68d:~/projects$ chmod o-w project_k.txt
researcher2@06ff4b1cc68d:~/projects$ chmod g-r project_m.txt
researcher2@06ff4b1cc68d:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 May 20 15:33 drafts
-rw-rw-r-- 1 researcher2 research_team  46 May 20 15:33 project_k.txt
-rw----- 1 researcher2 research_team  46 May 20 15:33 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 May 20 15:33 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 May 20 15:33 project_t.txt
researcher2@06ff4b1cc68d:~/projects$
```

Change file permissions on a hidden file

```
researcher2@06ff4b1cc68d:~/projects$ chmod u-w,g-w .project_x.txt
researcher2@06ff4b1cc68d:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 May 20 15:33 .
drwxr-xr-x 3 researcher2 research_team 4096 May 20 15:52 ..
-r----- 1 researcher2 research_team  46 May 20 15:33 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 May 20 15:33 drafts
-rw-rw-r-- 1 researcher2 research_team  46 May 20 15:33 project_k.txt
-rw----- 1 researcher2 research_team  46 May 20 15:33 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 May 20 15:33 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 May 20 15:33 project_t.txt
researcher2@06ff4b1cc68d:~/projects$ chmod g+r .project_x.txt
researcher2@06ff4b1cc68d:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 May 20 15:33 .
drwxr-xr-x 3 researcher2 research_team 4096 May 20 15:52 ..
-r----- 1 researcher2 research_team  46 May 20 15:33 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 May 20 15:33 drafts
-rw-rw-r-- 1 researcher2 research_team  46 May 20 15:33 project_k.txt
-rw----- 1 researcher2 research_team  46 May 20 15:33 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 May 20 15:33 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 May 20 15:33 project_t.txt
researcher2@06ff4b1cc68d:~/projects$ █
```

Change directory permissions

```
researcher2@06ff4b1cc68d:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 May 20 15:33 drafts
-rw-rw-r-- 1 researcher2 research_team  46 May 20 15:33 project_k.txt
-rw----- 1 researcher2 research_team  46 May 20 15:33 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 May 20 15:33 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 May 20 15:33 project_t.txt
researcher2@06ff4b1cc68d:~/projects$ chmod g-x drafts
researcher2@06ff4b1cc68d:~/projects$ ls -l
total 20
drwx----- 2 researcher2 research_team 4096 May 20 15:33 drafts
-rw-rw-r-- 1 researcher2 research_team  46 May 20 15:33 project_k.txt
-rw----- 1 researcher2 research_team  46 May 20 15:33 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 May 20 15:33 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 May 20 15:33 project_t.txt
researcher2@06ff4b1cc68d:~/projects$ █
```

Summary

File permissions and ownership is a necessary task to give privileges in order to keep a smooth business continuity, and to avoid attack surface.